

CONCLUSION

This study set out to examine the extent to which technological investment contributed to labour productivity growth across seven sub-sectors of the Brazilian services industry between 2012 and 2025. The empirical strategy followed a deliberate three-stage progression designed to ensure methodological rigour and to guard against the inferential pitfalls that commonly afflict time-series panel analyses. The first estimation, conducted on levels of the log-transformed variables, yielded a high coefficient of determination but failed every diagnostic test applied to its residuals: Augmented Dickey-Fuller unit root tests confirmed that the dependent variable was integrated of order one in all seven sectors, the Breusch-Godfrey statistic revealed severe serial autocorrelation, and the Durbin-Watson value of 0.534 signalled a classic spurious regression. Compounding this, the inclusion of quarterly time fixed effects induced perfect multicollinearity for the national-level technology investment variable, whose variation is exclusively temporal, rendering the first-stage coefficients economically meaningless. These results are not incidental: they illustrate, within a real Brazilian dataset, the precise mechanism through which regressions in non-stationary levels generate inflated fit statistics and unreliable inference, a methodological cautionary tale that the literature has documented since Granger and Newbold (1974). Correcting for these failures required transforming all series into first differences — equivalent to working with quarterly growth rates — which the ADF tests confirmed to be stationary, and adopting sector-clustered standard errors to account for residual heteroskedasticity, as corroborated by the Breusch-Pagan test. The removal of time dummies, justified by the elimination of the spurious stationarity problem and the need to preserve identifying variation in the technology investment regressor, produced VIF statistics close to unity, confirming that multicollinearity had been resolved. Only upon this methodologically sound foundation could the study's substantive findings be interpreted with confidence.

The central empirical finding of this research constitutes a formal, quantitative validation of the Solow Paradox (Solow, 1987; Brynjolfsson, Rock & Syverson, 2017) within the Brazilian services sector. In the baseline first-differences model — which captures contemporaneous and immediate short-run dynamics — the coefficient on the technology investment growth rate is statistically indistinguishable from zero across all specifications. This null result is not a failure of the empirical design; it is an economically meaningful signal. Firms that acquire digital equipment, software, or telecommunications infrastructure face a period of

organisational disruption during which resources are diverted from production to installation, staff training, and process adaptation. The productivity payoff is deferred, not absent. To test this hypothesis, the model was extended to include quarterly lags of the technology investment growth rate up to four periods. The results are unambiguous: lags zero through three remain statistically insignificant, but the fourth-quarter lag — corresponding to a twelve-month horizon — yields a coefficient of 0.054, significant at the five-percent level ($p = 0.014$). The economic interpretation is that, on average across the Brazilian services sector, a one-percent increase in the rate of growth of technology investment translates into a 5.4-percent uplift in labour productivity, but this effect materialises with a one-year delay. This finding is consistent with the implementation-lag explanation advanced by Brynjolfsson et al. (2017) as the dominant mechanism behind the modern productivity paradox, and with the endogenous growth logic of Romer (1990), whereby the productive absorption of new knowledge requires complementary investments in human capital and organisational capability that cannot be instantiated overnight. For Brazilian industrial policy, the implication is significant: the absence of short-run productivity returns to technology investment should not be interpreted as evidence that such investment is ineffective, but rather as a predictable characteristic of technological adoption cycles that demands patience and sustained commitment from both firms and policymakers.

The second major contribution of this study lies in its decomposition of the aggregate one-year technology effect into sector-specific estimates through an interaction model between the fourth-quarter technology lag and sector dummy variables. This heterogeneous-effects specification reveals that the services sector is far from a homogeneous absorber of technological investment: the productivity return to a one-percent increase in technology investment growth ranges from 18.3 percent in Information and Communication Technology services to 1.3 percent in Real Estate Activities, a differential of more than fourteen percentage points that carries both theoretical and policy significance. The “digital elite” — comprising ICT services and, to a lesser extent, Retail Trade (5.7 percent) and Transportation and Logistics (3.8 percent) — represents activities whose production functions are characterised by high task digitisability, rapid routinisation of cognitive processes, and well-established digital infrastructure. These are precisely the conditions under which Acemoglu and Restrepo (2018) predict that the productivity effect of automation and digitalisation will dominate the displacement effect, generating net gains in output per worker. Conversely, sectors at the base of the ranking — Real Estate Activities (1.3 percent) and Professional, Scientific and Technical

Activities (1.7 percent) — are characterised by intensive reliance on embodied tacit knowledge, face-to-face client interaction, and physical capital that is resistant to digital substitution. In these activities, the complementarity between digital investment and workforce capabilities, which Gal et al. (2019) identify as the decisive mediating variable in firm-level productivity analyses, appears weaker, either because the relevant skills are in short supply or because the tasks themselves present structural barriers to automation. The heterogeneity documented here is not merely a statistical artefact; it reflects the uneven nature of Brazil's digital transition and underscores the inadequacy of sector-agnostic technology policies.

Taken together, the findings of this study offer a coherent and empirically grounded narrative that resolves the apparent contradiction between the scale of Brazil's digital investment over the 2012–2025 period and the sluggishness of its measured labour productivity growth. Technology works — but it works slowly, and it works unevenly. The twelve-month maturation period identified here implies that short-run evaluations of technology policy will systematically understate its benefits, potentially leading to premature disinvestment or misplaced scepticism about the productive value of digitalisation. The sectoral heterogeneity findings imply that aggregate productivity statistics conceal a distributional story in which digitally mature sectors pull ahead while analogue-intensive activities stagnate, widening intra-service-sector productivity gaps and potentially reinforcing broader labour market inequalities of the kind discussed by Korinek, Schindler and Stiglitz (2021). Several limitations of the present study should be acknowledged. The technology investment proxy — constructed from Comex Stat import data for IT and telecommunications goods under NCM headings 8471, 8517, and 8542 — captures hardware and equipment acquisition but does not directly measure software investment, intangible assets, or the quality of managerial and organisational practices that the literature consistently identifies as critical complements to physical digitalisation. Similarly, the national-level variation in the technology investment series, necessitated by data availability, prevents identification of within-sector heterogeneity across firms of different sizes or ownership structures. Future research should seek to exploit firm-level administrative data, such as the Relatório Anual de Informações Sociais (RAIS) linked to tax and customs records, to test whether the one-year maturation lag and the sectoral productivity hierarchy documented here hold at finer levels of disaggregation, and to assess the extent to which workforce skill composition moderates the technology-productivity relationship in the Brazilian context.

REFERENCES

- ACEMOGLU, Daron; KONG, Dingwen; OZDAGLAR, Asuman. AI, Human Cognition and Knowledge Collapse. Cambridge, MA: National Bureau of Economic Research, 2026. (NBER Working Paper, n. 34910).
- ACEMOGLU, Daron; RESTREPO, Pascual. Artificial Intelligence, Automation and Work. Cambridge, MA: National Bureau of Economic Research, 2018. (NBER Working Paper, n. 24196).
- AGHION, Philippe; HOWITT, Peter. A Model of Growth Through Creative Destruction. *Econometrica*, v. 60, n. 2, p. 323–351, mar. 1992.
- BRYNJOLFSSON, Erik; ROCK, Daniel; SYVERSON, Chad. Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics. Cambridge, MA: National Bureau of Economic Research, 2017. (NBER Working Paper, n. 24001).
- GAL, Peter et al. Digitalisation and productivity: In search of the Holy Grail – Firm-level empirical evidence from EU countries. Paris: OECD Publishing, 2019. (OECD Economics Department Working Papers, n. 1548).
- GRANGER, Clive W. J.; NEWBOLD, Paul. Spurious Regressions in Econometrics. *Journal of Econometrics*, v. 2, n. 2, p. 111–120, 1974.
- KORINEK, Anton; SCHINDLER, Martin; STIGLITZ, Joseph E. Technological Progress, Artificial Intelligence, and Inclusive Growth. Washington, D.C.: International Monetary Fund, 2021. (IMF Working Paper, n. 21/166).
- ROMER, Paul M. Endogenous Technological Change. *Journal of Political Economy*, v. 98, n. 5, p. S71–S102, 1990.
- SIQUEIRA, Fernando de Faria; MAZALI, Rogério. Technological Change and its Economic Features in the Digital Era: Literature Review and Guidelines. Brasília: Universidade de Brasília, 2026.

SOLOW, Robert M. We'd Better Watch Out. New York Times Book Review, New York, 12 jul. 1987.
p. 36.